



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus

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College of Informatics and Computing Sciences

LABORATORY ACTIVITY 1
CS 413: ADVANCED SOFTWARE ENGINEERING

TEAM COMPOSITION.

No.	Member Name	Role
1	Royce Gabriel de Gala	Full-Stack
2	Justine Lance M. Pabalan	Full-Stack

PROJECT PROPOSAL

Proposed Project Title:	Flip - AI-Powered Flashcard & Quiz Builder
Problem Statement / Rationale:	Students spend significant time manually creating flashcards and quiz questions from lecture notes, textbooks, and PDFs, which cuts into actual study time. Existing tools like Quizlet require fully manual card creation or lock AI generation behind expensive plans, and don't integrate quiz-taking with adaptive review. A lightweight, studentpriced SaaS that turns raw notes into study-ready flashcards and quizzes would let students spend more time reviewing and less time formatting content.
Proposed Objectives:	• Allow users to create, organize, and study flashcard decks and quizzes through a clean web interface. • Use an LLM to auto-generate flashcards and quiz questions from pasted text or uploaded PDFs/notes. • Implement a freemium subscription model that gates advanced features (e.g., unlimited AI generation, PDF uploads, analytics) behind a paid tier. • Track user quiz performance over time to support spaced-repetition-style review.
Scope and Limitations:	The system will cover deck/quiz creation (manual and AI-generated), studying via flashcard and multiple-choice quiz modes, basic performance tracking, and subscription/paywall gating. It will not include real-time multiplayer quiz modes, mobile native apps (web-only, responsive design), or grading of free-text/essay answers. Full offline mode and OCR of handwritten notes are excluded since it's a website.
Target Users / Beneficiaries:	College and senior high school students preparing for exams; also useful for selflearners and professionals studying for certifications.



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COMPONENT WORKSHEET

Component Name	Reuse Type	Interface / Contract (in/out)	License	Build vs. Reuse Justification
Google Gemini API (LLM)	Black-box Reuse	In: raw text/notes + prompt template. Out: JSON array of flashcards/quiz questions.	Commercial/paid, usage-based pricing per token	Building an inhouse NLP model to generate quiz questions is far beyond semester scope; the API is well-documented and reliable at small scale, but ties cost directly to usage and requires rate limit handling.
Firebase Authentication	Black-box Reuse	In: email/password or Google OAuth credentials from the React client. Out: ID token verified server-side via Firebase Admin SDK, returning a decoded user object (UID, email) to Express route handlers.	Apache 2.0 (SDK) / Google Cloud ToS	Firebase owns password hashing, token issuance, and OAuth flow handling entirely, removing that implementation burden from the team; trade-off is coupling identity to Google's ecosystem and reliance on a third-party auth provider.
Stripe (Subscriptions/Billing)	Component Reuse	In: user ID + selected plan. Out: subscription status, webhook events on payment/renewal/cancellation.	Commercial (transaction-fee based), free to integrate	Handling PCIcompliant payments and recurring billing logic ourselves is impractical and risky; Stripe is the industry standard with strong docs, though it ties billing logic to their webhook events and adds a small transaction fee.



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Component Name	Reuse Type	Interface / Contract (in/out)	License	Build vs. Reuse Justification
Mongoose ODM + MongoDB Atlas	Component Reuse	In: Mongoose schema definitions/queries from the Express API. Out: structured deck, card, quiz, and user documents.	MIT License (Mongoose) / Server Side Public License (MongoDB)	A schema-based ODM keeps document structure consistent and reduces bugs versus raw MongoDB driver calls; Atlas's free tier is sufficient for a class project, though scaling later would require a paid cluster.
pdf-parse (PDF text extraction)	White-box Reuse	In: uploaded PDF file buffer (via Express fileupload middleware). Out: extracted plain text passed to the LLM for card generation.	MIT License	Writing a PDF parser from scratch is unnecessary; this library is lightweight and free and runs directly in the Node/Express backend, though it may need minor modification to handle multicolumn or scanned PDFs cleanly



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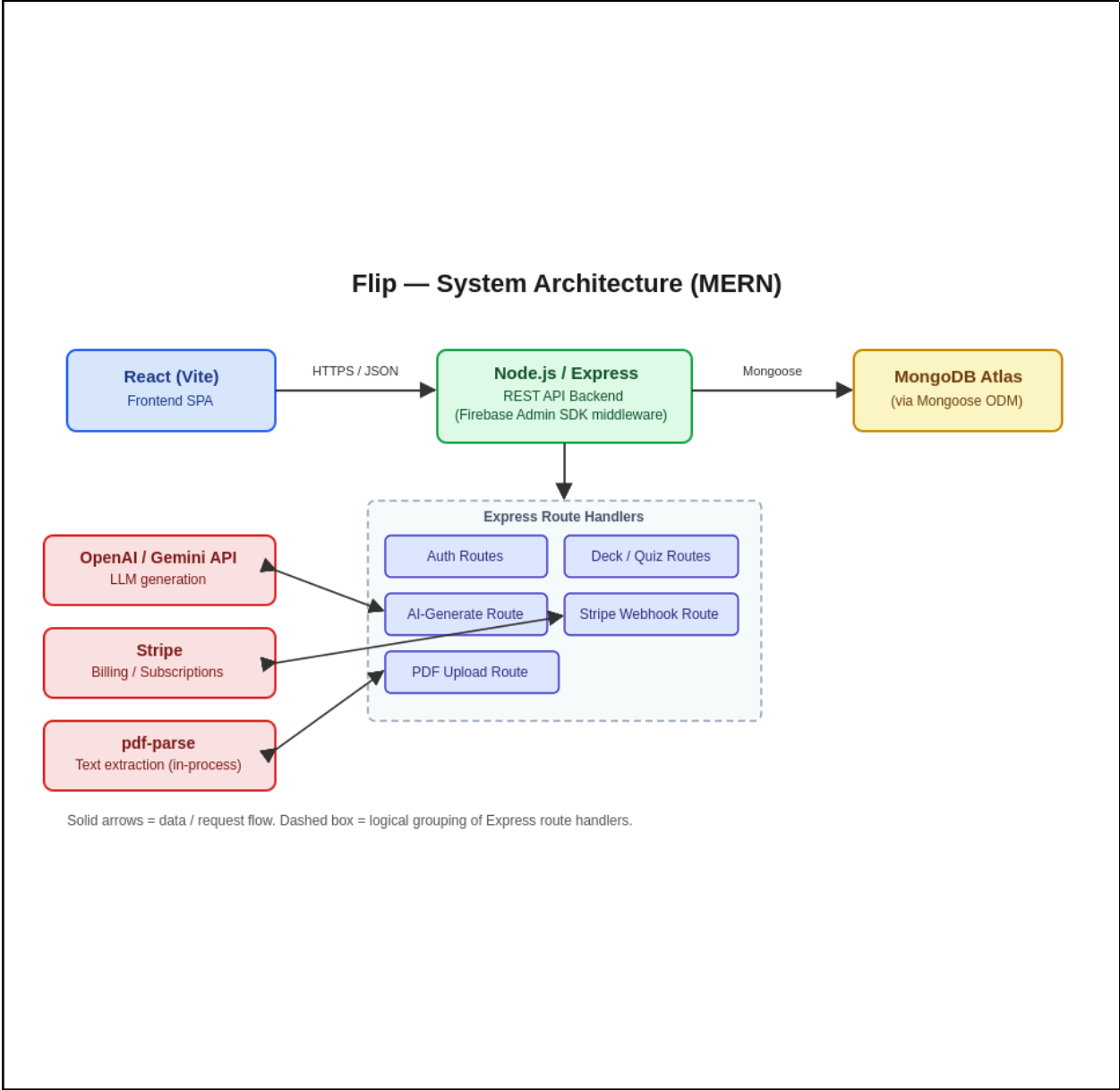
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PROPOSED SYSTEM COMPOSITION



Software Architectural Pattern	Client-server architecture using the MERN stack: a React (Vite) singlepage application on the frontend communicating with a Node.js/Express REST API backend, backed by MongoDB via Mongoose.
Integration Points:	• Firebase Admin SDK middleware sits at the Express API's entry layer, verifying the client-issued ID token on every protected route and attaching the decoded user (UID, email) to the request. • The LLM API is called from an Express route handler when a user submits notes/PDF text; results are validated and saved via Mongoose before being returned to the React frontend. • pdf-parse runs in an Express route (via multer file upload) to extract text before it is passed to the LLM.



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	• Stripe integrates via a webhook route in Express that updates the user's subscription status in MongoDB, which the API then checks to enable/disable premium features (e.g., unlimited AI generations, PDF uploads). • MongoDB/Mongoose is the central data layer connecting user accounts, decks, cards, quiz attempts, and subscription status, accessed exclusively through the Express API — the React frontend never talks to the database directly.
Known Design Risk	The LLM API's token-based pricing and rate limits mean heavy freetier usage could become costly or get throttled at scale, so free-tier users will have a capped number of monthly AI generations to control cost where this cap itself is an untested assumption about what "reasonable free usage" looks like and may need adjustment after real usage data.